



MongoDB and GenAI Glossary

A guide to key terms and concepts used when building, deploying, and evaluating generative AI applications with MongoDB.

Terms	Definitions
Algorithms	A finite sequence of well-defined, computer-implementable instructions.
Approximate Nearest Neighbor search (aNN)	Approximate Nearest Neighbor (aNN) search is a computational technique used to quickly find points in a dataset that are close to a given query point. aNN doesn't search every point in the graph, it uses the HNSW graph to search until it finds an approximate neighbor. This search is efficient, especially when you have a large dataset, but there's a slight chance of having an outlier in your results.
Artificial Intelligence	A field in computer science that trains computers to simulate human intelligence.
Atlas Vector Search	Atlas Vector Search is a feature in MongoDB Atlas that allows users to perform fast and efficient similarity searches on high-dimensional data by using vector representations. It enables semantic search capabilities by comparing vectors to find the most relevant or similar items to a given query.
Chunk overlap	The number of overlapping tokens between two adjacent chunks. This overlap will create duplicate data across chunks. Overlap can help preserve context between chunks and improve your results.

Chunk size	The maximum number of tokens contained in each chunk.
Chunking	The process of breaking down large amounts of information or data into smaller, more manageable pieces, or "chunks." These chunks are designed to be easily processed, stored, and retrieved by the AI model.
Chunking Strategy	Chunking strategy involves planning and implementing the optimal way to divide information into chunks to maximize the performance of the AI system. Chunking strategy takes into account the nature of the data, the specific requirements of the task, and the capabilities of the AI model. The goal is to create chunks that are coherent, contextually relevant, and of an appropriate size to be processed effectively by the model.
Context Precision	Context Precision represents the ability of a retriever to order retrieved items by relevance to the ground truth.
Context Recall	Context Recall represents how aligned the context retrieved is to the ground truth.
Context window	The context window is the amount of data that an LLM can consider at one time. The size of an LLM's context window is measured in tokens.
Cosine similarity	Cosine similarity is a metric that uses the angle between two vectors to determine the similarity between those vectors. Note, that cosine does not take magnitude into account, and you can't use zero magnitude vectors with cosine. To measure cosine similarity, it is recommended to normalize the vectors and use dotProduct instead.

Coupled architecture	The default MongoDB Atlas model of running your database and search workloads on the same node.
Data Ingestion Pipeline	A series of steps that involve importing, organizing, and transforming data from various sources into the destination system. This process ensures that the data is high quality by the time it is retrieved by a RAG application.
Decoder models	Decoder models use only the decoder of a Transformer model. They take embeddings and generate an output based on the context provided by the embeddings. Pretraining of these models usually revolves around predicting the next word in a sentence.
Deep Learning	Deep learning is a subset of machine learning that involves algorithms inspired by the structure and function of the brain, known as artificial neural networks.
Dense vectors	A dense vector is a numeric representation of data where most or all the dimensions contain non-zero values. Dense vectors are commonly used in semantic searches and machine learning applications to represent complex, abstract features, such as the meaning of words, phrases, or entire documents, in a continuous, high-dimensional space. Dense vectors capture semantic relationships—the deeper meaning or similarity between data points. For example, in Atlas Vector Search, dense vectors allow the system to find documents or phrases that are conceptually similar to a query, even if they don't share exact words.

Dimensions	The dimensions of a vector embedding represent the features or attributes of the data item in a high-dimensional space. Each dimension corresponds to a particular aspect or characteristic of the data, and the value in each dimension represents the strength or presence of that characteristic.
Document specific splitters	Document Specific splitters use separators that are unique to the document type to create chunks.
dotProduct similarity	Dot product similarity is a measure of similarity between two vectors in a vector space, calculated as the sum of the products of their corresponding components. To use dotProduct, the vector must be normalized to unit length at index- and query-time.
Embedding	An array of numerical data, or vectors, representing a piece of information, such as text, images, audio, video, etc.
Embedding Model	A deep learning model used to transform data into vectors, also known as embeddings.
Encoder models	Encoder models only use the encoder from the Transformer model. Encoders take in an input and return embeddings.
Euclidean similarity	Euclidean similarity uses the distance between two vectors in a multidimensional space to calculate the similarity between those vectors. The underlying formula is derived from the Pythagorean theorem and generalized for any number of dimensions.

Faithfulness	A measure of generation performance intended to quantify the factual correctness of a generated answer compared to the retrieved context. I.e. how close is the answer to the chunks provided by the retriever.
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Feedforward	Feedforward neural networks are the simplest type of neural network. The connections between nodes are linear and move forward. Feedforward neural networks are widely used for tasks like classification.
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Generative AI	GenAI leverages large language models to create text, images, video, and audio based on a given prompt.
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Generator	In the context of RAG, the main purpose of the generator is to come up with a response to the query based on the context provided by the retriever. Two main components of the generator are the prompt and the generative model which generates the answer.
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Ground Truth	A factually correct statement provided in a dataset to compare with LLM responses.
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Hallucinations	Situations where AI generates coherent but inaccurate information. Hallucinations can erode trust in the application and are often difficult to identify and address.
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Hierarchical Navigable Small World (HNSW) graphs

HNSW graphs plot multiple points with long and short links, similar to an NSW graph. But unlike NSW graphs which are on one plane, HNSW graphs are made up of multiple layers, like a skip list.

Layer zero contains all the plotted points with shorter links. As we move up in the HNSW graph, there are fewer points and the links become longer. This creates a simpler view of the layer below it. The top layer will have a few points with the greatest distance between them. MongoDB Vector Search uses the Hierarchical Navigable Small Worlds algorithm for efficient `$vectorSearch` queries.

Hybrid Search

Hybrid Search is a feature that combines traditional full text search and semantic search to produce comprehensive and contextually relevant search results.

Knowledge Base

In the context of RAG, a knowledge base is a repository that stores and organizes data, so it's easy to retrieve information that models use to generate responses.

Langchain

An AI framework that connects the components of a RAG system, such as the retriever and answer generator, using chains.

Large Language Model (LLM)

A type of artificial intelligence (AI) that specializes in processing text data and generating human-like text.

These models can recognize complex patterns, understand context, make predictions, and more.

Machine Learning

A field in AI that creates algorithms and models to help computers digest vast amounts of data, which can be structured or unstructured.

Model	A model is a mathematical representation or algorithm that is trained on data to recognize patterns and make predictions or decisions. The algorithm dictates how the learning process happens, and the outcome of this learning process is a model.
mongod	mongod is the primary daemon process for the MongoDB system. It handles data requests, manages data access, and performs background management operations. mongod takes care of storing and managing the core data in your MongoDB database.
mongot	The mongot process creates Atlas Search indexes based on the rules in the index definition for the collection. It also monitors change streams for the current state of the documents and indexes for the collections for which you defined Atlas Search indexes. It also processes Atlas Search queries and returns the document IDs and other search metadata for the matching documents to mongod, which then does a full document lookup and returns the results to the client.
MTEB	An industry-standard benchmark dataset generally used to evaluate embedding models' performance.
multi-modal	In the context of AI, multi-modal refers to LLMs that are capable of processing data beyond text, such as images, audio, and video.
Natural Language Processing (NLP)	NLP gives computers the ability to understand and work with human languages.

Navigable Small World (NSW) graphs

NSW graphs are a type of proximity graph used to find similarities. NSW graphs are made up of vertices with short and long links. Each vertex, or node, is a data point in a highly dimensional space. Each link, or edge, represents the similarity between the two points.

Neural networks

A computational model inspired by the structure and functional aspects of the brain.

A neural network is used to model complex relationships between inputs and outputs and to find patterns in data.

numCandidates

numCandidates refers to the number of initial entry points, or starting vectors, used during the search process in an HNSW (Hierarchical Navigable Small World) graph. In the context of approximate nearest neighbor search, the algorithm begins searching from a random point on the top layer of the graph and continues refining the search as it moves down through layers.

Prompt

In the context of RAG, a prompt consists of the user's query, the relevant chunks of data, and metadata, which are provided as input to a LLM.

Ragas

Ragas is an open-source framework for evaluating RAG applications that includes support for metrics such as context precision and recall and integrates with frameworks like LangChain and LlamaIndex.

Reciprocal Rank Fusion

Reciprocal Rank Fusion (RRF) is an information retrieval technique used to combine the ranked results from multiple search engines or algorithms into a single, more accurate ranking. It works by giving higher importance to results that are ranked highly by any of the individual ranking systems while ensuring that results ranked lower also get some consideration.

Recurrent Neural Networks (RNNs)

Similar to a feedforward network, RNNs move forward through nodes, but RNNs have the ability to loop backwards. This allows RNNs to not only leverage the data it's been given, but it can also use what it has learned from the previous input.

Recursive text splitter

A technique when chunking source data that preserves semantic relationships in text by using separators like line breaks to recursively divide text into smaller and smaller chunks.

Retrieval Augmented Generation (RAG)

An AI framework that aims to improve the quality of pre-trained LLMs by providing them with relevant contextual, up-to-date data retrieved from a knowledge base. RAG works by combining an embedding model and a generative model, to retrieve relevant information from a dataset and provide detailed responses based on that information.

Retrieval System

A system designed to search, find, and rank documents or information from a knowledge library, such as a database, based on a user query.

Retriever

In the context of RAG, the main job of the retriever is to take the user's query and find chunks of data that relate to the query.

Sanitize	In the context of RAG, sanitizing is the process of cleaning, preparing, and formatting data so that it is usable, accurate, and free of any potentially harmful or redundant elements. Sanitized data can assist with the retrieval of relevant information and enhance the quality of a generated response.
Search Nodes	Search nodes are dedicated components or servers that are responsible for handling search indexing and query functionalities. These nodes allow for workload isolation without data isolation, and ~90% of their RAM allocation can be used to store the vector data and indexes in memory, with the leftover being used for the JVM.
Search Page Faults	Search Page Faults is the average rate of page faults on a process per second over the selected sample period.
Semantic search	Semantic search interprets the meaning of words and phrases. The results of a semantic search will return content matching the meaning of a query, as opposed to content that literally matches words in the query.
Semantic splitting	A strategy when chunking source data that uses vector embeddings to split text based on semantic similarity.
Sequence to sequence	A model type that uses both encoders and decoders. Sequence to sequence models excel at tasks that involve taking an input and generating a sentence based on that input.
Skip list	A skip list is a data structure that allows fast search, insertion, and deletion operations within an ordered sequence of elements. It is similar to a linked list, but it includes multiple layers of linked lists to "skip" over large portions of the data, speeding up access to elements.

Sparse Vectors	A sparse vector is a type of numeric representation used primarily in text searches, where most of its dimensions are zeros, and only a few dimensions contain non-zero values. These vectors are designed to capture the presence or absence of specific features—such as words in a document—while ignoring irrelevant or missing features.
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System CPU	System CPU represents the utilization of the underlying CPU instances. The System CPU helps determine if there is bottlenecking on serving requests.
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System Memory	System Memory represents the number of bytes of system memory available for running new applications without swapping. This can be used to find out if an index is loaded into filesystem cache.
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Transformer Model	A transformer model is a neural network designed to learn context and meaning by analyzing relationships in sequential data, such as the words in a sentence. Transformer models use a set of mathematical techniques, known as attention or self-attention, to identify how elements in a sequence influence and depend on each other. The ability to focus on different parts of the input sequence allows the model to understand and generate complex data with high accuracy. The transformer model consists of encoders and decoders. Both the encoder and decoder have multiple layers inside that are used to calculate predictions.
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Vectors	Numerical representations of unstructured data like text, images, and audio.
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